EthnoHack 2026

Data Hackathon on Ethnopharmacology & Health

2–8 November 2026

100% Online · CET

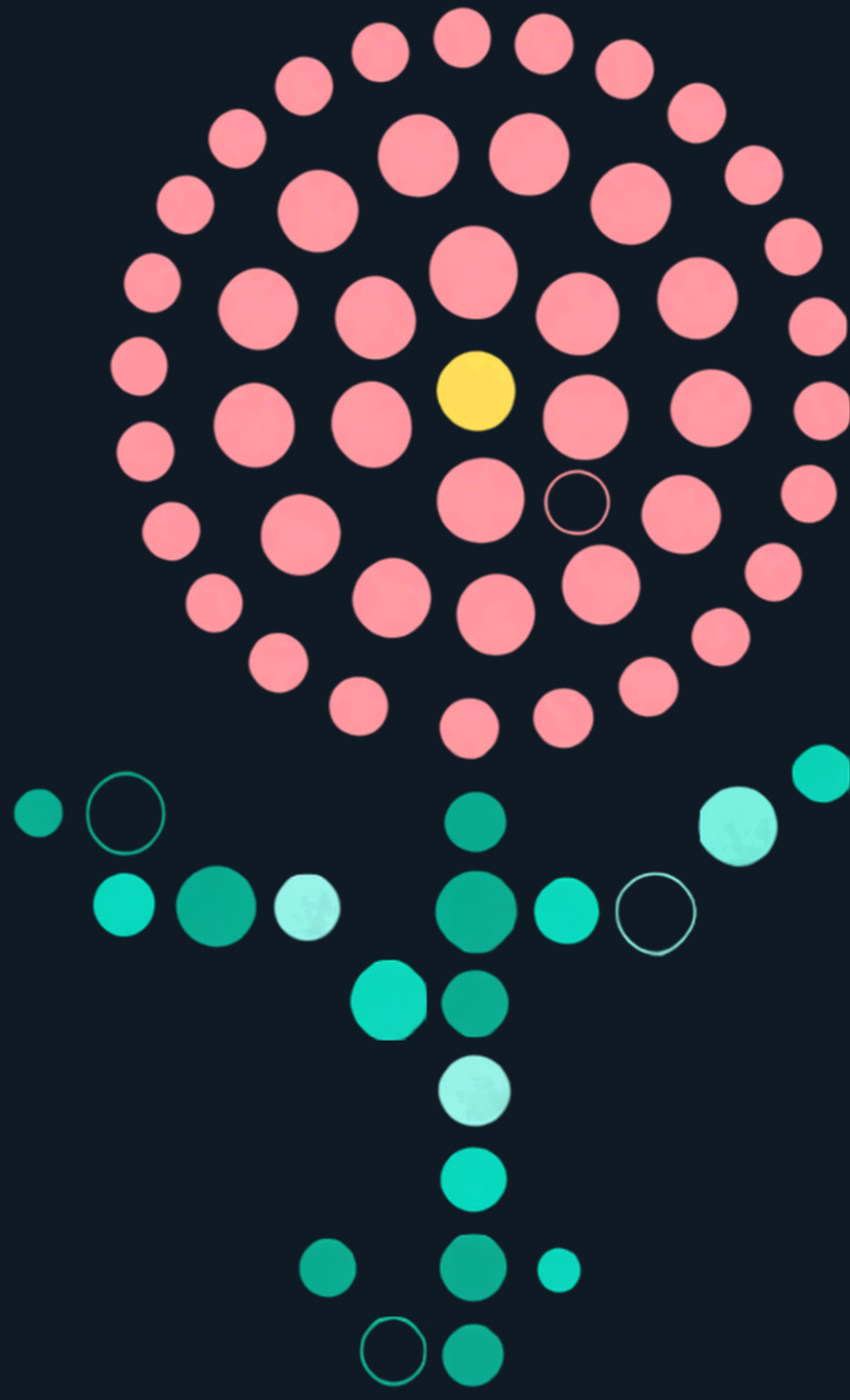
ETHCSTWIN

HELLENIC REPUBLIC
National and Kapodistrian University of Athens

European Commission

Horizon 2020
European Union funding
for Research & Innovation

Questions???
Write in the chat.



WHERE ETHNOHACK COMES FROM

ETHCSTWIN: ethnopharmacology meets computer science

A Horizon Europe project twinning the Institute for Ethnopharmacological Studies & Phytotherapy (IESP, University of Athens) with computer science, so traditional plant knowledge can be studied with AI and data.

Funded by Horizon Europe

June 2024 – May 2027

5 partners · 4 countries

Network

Link the Balkans, the Mediterranean and Central Europe.

Data

Bring manuscripts, folklore and field studies into one database.

Tools

Build methods to analyse large datasets and restore heritage.

Skills

Transfer know-how in AI, machine learning and data mining.

THE CONSORTIUM

Five partners, four countries

GREECE	**IESP, National & Kapodistrian University of Athens**	Traditional plant use and bioactivity
GERMANY	**Fraunhofer SCAI, Department of Bioinformatics**	Text mining, knowledge modelling, biomedical AI
GERMANY	**Pangea Botanica (PangeAI)**	Plant-use data for drug discovery
ITALY	**University of Gastronomic Sciences, Pollenzo**	Food systems and medicinal home foods
KOSOVO	**University of Prishtina, Department of Biology**	Balkan ethnobotany fieldwork, plant diversity

WHY WE'RE DOING THIS

Centuries of plant knowledge, waiting to be connected

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It's scattered

Manuscripts, folklore and field notes, in many languages and with old plant names.

It's at risk

Living traditions fade, and the plants themselves face pressure from harvesting.

Data can unlock it

NLP, cheminformatics and maps link old remedies to compounds, targets and care.

No single discipline can do this alone. That's why EthnoHack puts coders and plant scientists on the same team.

THIS WEEK'S MISSION

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Turn traditional knowledge into evidence for drug discovery.

THE PROGRAM

One week, three phases

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MONDAY · TODAY

Kick-off

10:00 CET Welcome
10:30 CET Track presentations
11:00 CET Team formation
13:00 CET Hacking begins

MONDAY – SATURDAY

Build

Hack in your own hours, any time zone. Mentoring and Q&A run all week on Discord.

SUNDAY 8 NOV

Pitch & win

10:00 CET Submission deadline
1 week judge deliberation

SUNDAY 15 NOV

Winners Announcement

TEAMS

Mixed teams build better projects

1

Join as a team

Already have a team? Great. On your own? You'll find one today.

2

Team formation, 11:30

Pitch an idea or join one, by track, live and in #track-*-find-a-team.

3

Register your team

Create your devpost project page, add teammates and choose your track. Send message to #request-team-channel for discord private chat.

Teams of [2–5]. Mix disciplines: a coder, a scientist and a domain expert beat three of the same.

DISCORD

Where the week happens

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#announcements

Schedule changes, deadlines and live links. Read-only.

#track-1 … #track-4

Track questions, datasets and talk with the track lead.

#faq-and-resources

Find the most popular questions answered and the resources we provide.

#event-questions

Tech, Devpost or rules questions for the organisers.

#track-*-find-a-team

Post your skills and idea; find teammates.

#request-team-channel

Once your team has formed, you can request a private workspace for your team.

#mentor-room or #find-a-mentor

Request help from a mentor by field.

Your team channel

Private text and voice space, once your team is set.

MENTORS

Experts across every track

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Evanthia Dina
Ethnobotany

Vasilis Kaintas
Pharmacognosy

Avni Hajdari
Ethnobotany

Konstantia Graikou
Natural Products Chemistry

Ioanna Tsitsou
Historian

David Buchmann
Bioinformatics

Marc Jacobs
Computer Scientist

Vijay Ingalalli
Data Science

Michael Chenos
DevOps

Christian Ebeling
Bioinformatics

Mihrije Kadriu
Computer Scientist

M Rumman Aslam
Programming

Anshulika Saxena
Bioinformatics

Manika Theodora
Computer Scientist

George Baltatzis
Biologist

MENTORS

How to get a mentor’s help

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Ask any time

Post in #mentor-room with your team, track and question. Tag the field you need.

Join office hours

Live voice sessions every day. Schedule pinned in #announcements.

They guide, you build

Mentors help you think, not write your code. Come with a specific question or idea.

Expect a reply within [12 hours]. Mentors are across time zones — ask early, not at 9:55 on Saturday.

DELIVERABLES

Submit four things by Sunday, 10:00 CET

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Devpost page
Title, description, team and track.

GitHub repo
Code, notebooks and a clear README.

Pitch video
10 minutes: problem, approach, results, impact, demo.

Slide deck
At most 10 slides.

Judged on scientific relevance, innovation, technical quality, data science, feasibility, impact and presentation.

PRIZES

A first prize for every track.

Each of the four tracks has its own first prize. What's inside?

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TRACK 01



1st Prize

Amount to be revealed

Traditional Knowledge Mining

TRACK 02



1st Prize

Amount to be revealed

Natural Products for Drug Discovery

TRACK 03



1st Prize

Amount to be revealed

Biodiversity & Sustainability

TRACK 04



1st Prize

Amount to be revealed

Digital Health & Phytotherapy

PRIZES

A first prize for every track.

Each of the four tracks has its own first prize: €2,000 per winning team.

ETHCSTWIN

TRACK 01



1st Prize

€2,000

Traditional Knowledge Mining

TRACK 02



1st Prize

€2,000

Natural Products for Drug Discovery

TRACK 03



1st Prize

€2,000

Biodiversity & Sustainability

TRACK 04



1st Prize

€2,000

Digital Health & Phytotherapy

PRIZES

Special category awards

Open to every team, whatever the track. Chosen by the jury alongside the track prizes.

Best AI Solution

Most innovative use of machine learning.

Best Student Team

Top team made up entirely of students (BSc, MSc, PhD).

Best Scientific Rigor

Most thorough design and data validation.

Most Innovative Application

Breakthrough solution to an unexpected real-world problem.

Best Community Impact

Strongest direct benefit to local communities.

Best Collaboration

Outstanding teamwork across disciplines or institutions.

SPONSORS

Thank you to our sponsors

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IN-KIND

Anakin

Web data as an API

Aitho

Pitch rehearsal

COMMUNITY





IN-KIND SPONSORS

Free tools for every participant

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Anakin

Turns any website into a structured API, so you can pull live web data without writing scrapers.

For you: one free month of the $20 plan.

Winning team: the $100 plan free for a month, then 50% off for 3 months.

Aitho

A rehearsal tool for presenters: speak your talk and the script follows your voice.

For you: Presenter Pro free through 31 December.

Perfect for practising your 10-minute pitch.

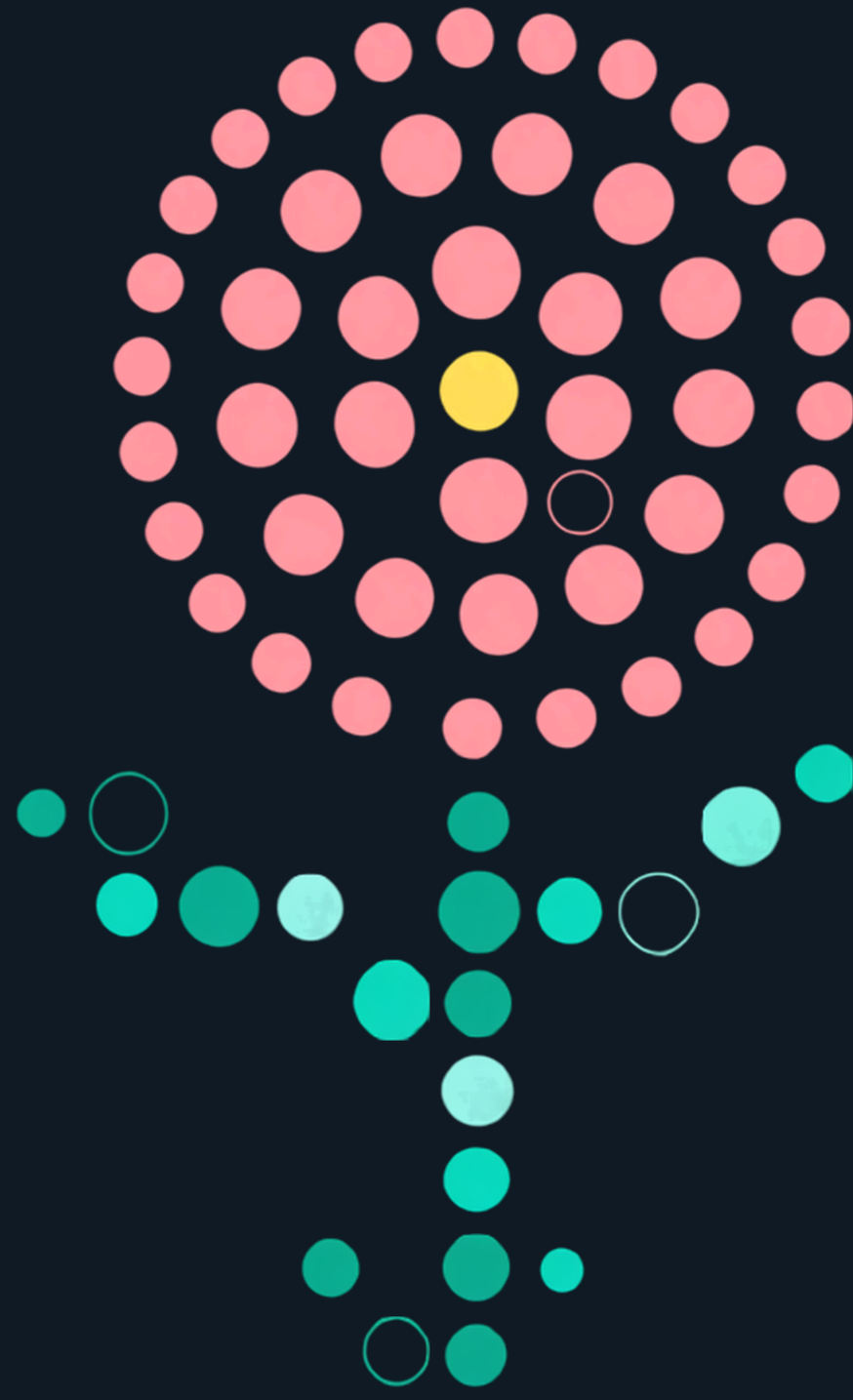
Sign-up links are in the faq-and-resourses on Discord. You sign up yourself; we share no emails with sponsors.

CHALLENGE PRESENTATIONS

Meet the tracks

Four open challenges. The directions are inspiration — your own ideas are welcome.

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CHALLENGE TRACKS

Choose one of four tracks

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TRACK 01

Traditional Knowledge Mining

NLP and AI on historical, folklore and ethnobotanical texts.

Lead: Marc Jacobs

TRACK 02

Natural Products for Drug Discovery

ML and cheminformatics to find plant compounds for disease targets.

Lead: Antigoni Cheilari

TRACK 03

Biodiversity & Sustainability

Geospatial and ecological data for conservation and sustainable harvest.

Lead: Evanthia Dina

TRACK 04

Digital Health & Phytotherapy

Apps and decision tools linking phytotherapy evidence to care.

Lead: Nektarios Aligiannis

TRACK 01

Traditional Knowledge Mining

Lead: Marc Jacobs

What did centuries of texts know about medicinal plants — and does modern science agree?

DIRECTIONS TO SPARK IDEAS

- Plant & disease names in old texts
- Themes across the literature
- Ancient diseases, modern terms
- A plant’s reputation over time
- Knowledge graphs

OPEN DATA SUGGESTIONS TO EXPLORE

Biodiversity Heritage Library · Wellcome Collection · Dioscorides & Avicenna texts · PubMed · Europe PMC · ICD11 · ICPC3 ·OpenAlex · Kaggle/Hugging face datasets

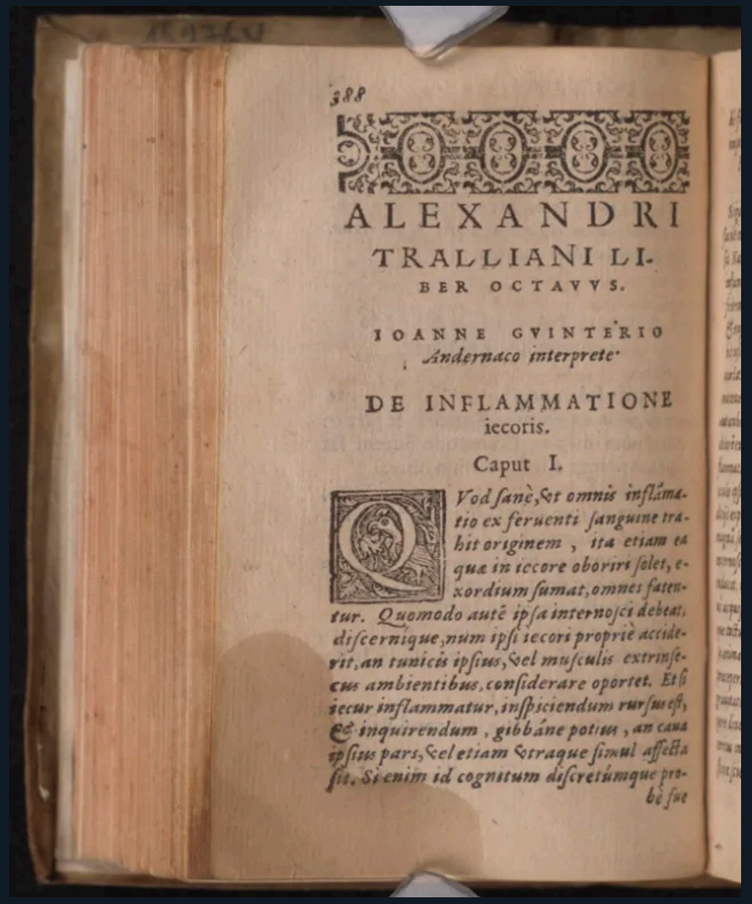
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Inspiration only: your idea, your approach. Combine tracks or go somewhere new.

TRACK 01

Traditional Knowledge Mining

Challenges: digitize - translate - interpretate - compare


388

ALEXANDRI
TRALLIANI LI-
BER OCTAVVS.

IOANNE GVINTERIO
Andernaco interprete.

DE INFLAMMATIONE
iecoris.

Caput I.

Quod sanè, & omnis inflamatio ex feruenti sanguine trahit originem, ita etiam ea quæ in iecore oboriri solet, exordium sumat, omnes fatentur. Quomodo autē ipsa internosci debeat, discernique, num ipsi iecori propriè acciderit, an tunicis ipsius, vel musculis extrinsecus ambientibus, considerare oportet. Etsi iecur inflammatur, inspiciendum rursus est, & inquirendum, gibbáne potius, an cava ipsius pars, vel etiam vtraque simul affecta sit. Si enim id cognitum discretúmque prius fue

frogs, onion, alcyonium, deadly carrot, spurge, mustard, and others

er ones and enhancing — due to the evacuation of h

recovering from di — ng blood alone is suffici

made by ladanum

adanum in myrtle — honey, and anoint the

call adianton) at half t — with the mixture with m

e mildly warming and — y of nard oil or henn

medy]

ssom of the anemone with oil and anoint [the area].

deadly carrot

simple compound

Adams: *Thapsia Garganica, Thapsia Asclepium*; **André**, LSJ: *Thapsia garganica*; **Puschmann:** *Thapsia* spp.

Thapsia, Thapsia Asclepium, Thapsia Garganica, Thapsia garganica, deadly carrot, θαψια

ETHCSTWIN:VEGETAL/FA1F4A23-0F34-38AC-AC62-018BC2348EC4

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κεφ. γ′. Πρὸς μέλανσιν τριχῶν.

Κηκίδων, ἀκακίας, λεπίδος σιδήρου, χαλκάνθου, στυπτηρίας ἑξ ἴσου λαβὼν ἀπόβρεξον οὔρῳ παιδὸς ἀφθόρου ἡμέραν μίαν, εἶτα σμήξας τὰς τρίχας χρῖσον καὶ δῆσον τὴν κεφαλὴν ἐφ᾽ ἡμέρας τρεῖς· πεπείραται.

Ἄλλο·

Καρύων χλωρῶν τοῦ φλοιοῦ .. οὐγγ. γ′ πρίνου καρποῦ ..... οὐγγ. ς′ οἴνου μέλανος ..... ξεστ. γ′. ἕψε, ἕως εἰς τρίτον ἔλθῃ, καὶ τὸ λοιπὸν ἐκθλίψας ἀνάκοπτε ἐλαίῳ μυρσίνῳ ξεστ. α′· ἡ χρῆσις καθ᾽ ἡμέραν.

Chapter 3. For darkening the hair

Take equal parts of oak galls, acacia, iron flake, blue vitriol, and alum. Soak them for one day in the urine of an uncorrupted boy; then, after washing the hair, anoint it with the mixture and bind up the head for three days. This has been tried.

Another [remedy]. Three ounces of green walnut shells; six ounces of the fruit of the holm oak; three xestai of dark wine. Boil until [the mixture] is reduced to one third, then press out the remainder and mix it with one xestēs of myrtle oil. Use it daily.

Another [remedy]

Disease	Simple Drug	Simple Drug Part	Compound	Dosage Form	Ingredients	Source
sleepiness			oxyrrhodinon		rose oil; vinegar	A mixture of rose oil and vinegar is particularly effective in this regard, because it strengthens the head, especially at the onset of the illness, so that it later no longer readily absorbs the many vapors that rise to it.
sleepiness			<no name>		rose oil; vinegar; sulphur-wort	one should mix rose oil and vinegar and cook it with substances that have a thinning effect, such as sulphur-wort, castoreum, pennyroyal , mint or thyme

TRACK 02

Natural Products for Drug Discovery

Lead: Antigoni Cheilari

Among hundreds of thousands of plant compounds, which deserve a closer look?

DIRECTIONS TO SPARK IDEAS

- Bioactivity prediction for extracts, fractions and pure compounds
- Drug-likeness filtering
- Virtual screening
- Mapping chemical space
- Plants bridging old and new medicine
- Plant–compound–target graphs
- Synergy

OPEN DATA SUGGESTIONS TO EXPLORE

COCONUT · NPASS · ChEMBL · PubChem · KNApSAcK · UniProt · PDB · DrugBank · KEGG · Kaggle/Hugging face datasets

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Inspiration only: your idea, your approach. Combine tracks or go somewhere new.

TRACK 03

Biodiversity & Sustainability

Lead: Evanthia Dina

Where do medicinal plants grow, which are at risk, and how do we keep them?

DIRECTIONS TO SPARK IDEAS

- Diversity hotspots and risk
- Climate-driven range shifts
- Sustainable substitutes
- An interactive plant atlas
- Conservation priorities
- Illegal-trade signals
- Nagoya Protocol
- Acess & Benefit sharing

OPEN DATA SUGGESTIONS TO EXPLORE

GBIF · World Flora Online · Plants of the World Online · Inaturalist · IPNI · IUCN Red List · TRY · Global Forest Watch · Kaggle/Hugging face datasets

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Inspiration only: your idea, your approach. Combine tracks or go somewhere new.

TRACK 04

Digital Health & Phytotherapy

Lead: Nektarios Aligiannis

How do we make phytotherapy knowledge usable, safe and evidence-based to be incorporated in Health Systems?

DIRECTIONS TO SPARK IDEAS

- Herbal product recommender
- Herb–drug interactions
- Evidence explained for patients
- Safety signal detection
- Monographs to clinical terms
- Health providers education

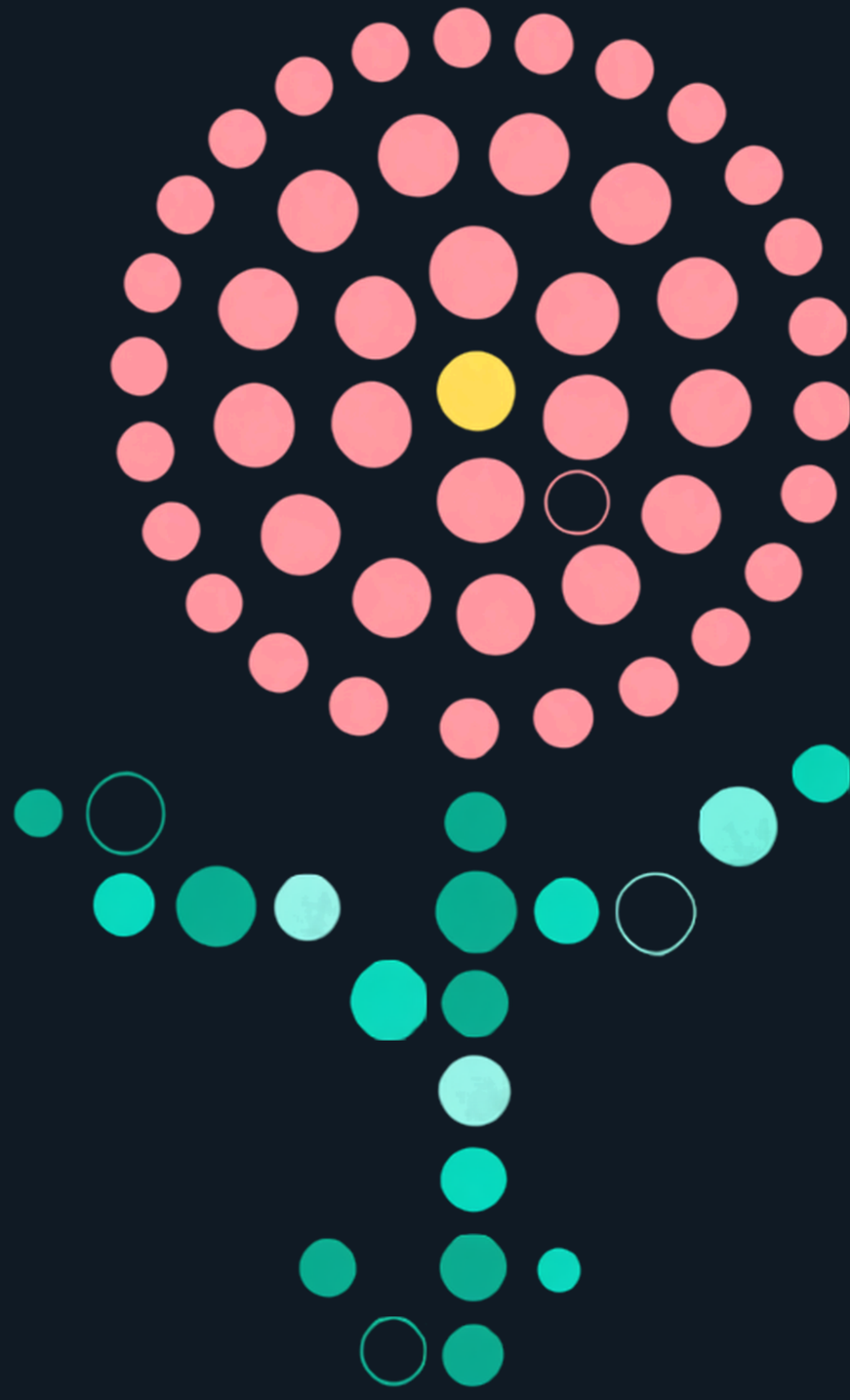
OPEN DATA SUGGESTIONS TO EXPLORE

EMA herbal monographs · WHO monographs · ICD11 · ICPC3 · ESCOP · DrugBank · OpenFDA · RxNorm · Kaggle/Hugging face datasets

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Inspiration only: your idea, your approach. Combine tracks or go somewhere new.

Questions???
Write in the chat.



NEXT

Find your team. Let’s build.

Team formation starts now in #team-formation. Hacking begins at 13:00.

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